

# **TECHNICAL REPORT:**

## **Secure Infrastructure & DNS Hardening**

**Project:** h4ckai.uk Home Lab Security

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**Date:** March 15, 2026

**Status:** IMPLEMENTED / VERIFIE

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# 1. Executive Summary

This report documents the comprehensive security hardening of the `h4ckai.uk` network infrastructure. The project focused on two critical security pillars:

1. **DNS Privacy:** Eliminating plain-text DNS leaks to external entities.
2. **Internal Transport Security:** Transitioning from unencrypted HTTP to mandatory HTTPS for all internal administrative panels.

By implementing centralized SSL/TLS termination, we successfully eliminated the "Not Secure" warnings across the local area network (LAN), ensuring that all internal management traffic (including credentials) is encrypted and verified against local trust authorities. This project was executed as part of advanced practical studies in Network Security and Ethical Hacking methodologies.

## 2. Infrastructure Components

- **Hypervisor:** Proxmox VE
  - **DNS Server:** Technitium DNS (Self-hosted)
  - **Reverse Proxy:** Nginx Proxy Manager (NPM)
  - **Security Protocols:** DNS-over-TLS (DoT), TLS 1.3, SSL (Let's Encrypt)
-

### 3. Implementation Details & Evidence

#### 3.1 Authoritative DNS Management

**Description:** Centralized management of the local zone for `h4ckai.uk`.

- **Key Action:** Implemented A-records and Wildcard steering to route internal traffic efficiently.
- **Security Benefit:** Reduces internal reconnaissance surface and centralizes traffic control.

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h4ckai.uk

PrimaryEnabled

Add RecordDisable ZoneDelete ZoneOptionsPermissionsDNSSEC

Nameabc or a\* or \*b\* or a?c

Type

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1-7 (7) of 7 records (page 1 of 1)

| # | Name    | Type | TTL        | Data                                                                                                                                                                                                                                                                                                                                                                                                             |
|---|---------|------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | @       | NS   | 14400 (4h) | <div>Name Server: technitiumdns</div> <div>Last Used: 0001-01-01 00:00:00 (never)</div> <div>Last Modified: 2026-03-14 22:15:11 (21 hours ago)</div>                                                                                                                                                                                                                                                             |
| 2 | @       | SOA  | 900 (15m)  | <div>Primary Name Server: technitiumdns</div> <div>Responsible Person: hostadmin@h4ckai.uk</div> <div>Serial: 13</div> <div>Refresh: 900 (15m)</div> <div>Retry: 300 (5m)</div> <div>Expire: 604800 (1w)</div> <div>Minimum: 900 (15m)</div> <div>Use Serial Date Scheme: false</div> <div>Last Used: 2026-03-15 19:33:01 (a few seconds ago)</div> <div>Last Modified: 2026-03-14 22:15:11 (21 hours ago)</div> |
| 3 | *       | A    | 3600 (1h)  | <div>Last Used: 2026-03-15 12:12:01 (7 hours ago)</div> <div>Last Modified: 2026-03-14 22:17:18 (21 hours ago)</div>                                                                                                                                                                                                                                                                                             |
| 4 | dns     | A    | 3600 (1h)  | <div>Last Used: 2026-03-15 19:33:01 (a few seconds ago)</div> <div>Last Modified: 2026-03-14 22:17:36 (21 hours ago)</div>                                                                                                                                                                                                                                                                                       |
| 5 | grafana | A    | 3600 (1h)  | <div>Last Used: 2026-03-15 12:01:15 (8 hours ago)</div> <div>Last Modified: 2026-03-15 11:50:53 (8 hours ago)</div>                                                                                                                                                                                                                                                                                              |
| 6 | pve     | A    | 3600 (1h)  | <div>Last Used: 2026-03-15 11:54:17 (8 hours ago)</div> <div>Last Modified: 2026-03-15 11:53:43 (8 hours ago)</div>                                                                                                                                                                                                                                                                                              |
| 7 | wazuh   | A    | 3600 (1h)  | <div>Last Used: 2026-03-15 11:51:31 (8 hours ago)</div> <div>Last Modified: 2026-03-15 11:51:08 (8 hours ago)</div>                                                                                                                                                                                                                                                                                              |

1-7 (7) of 7 records (page 1 of 1)

## 3.2 DNS-over-TLS (DoT) Hardening

**Description:** Transitioning from standard UDP/53 to encrypted DNS.

- **Protocol:** DNS-over-TLS (Port 853)
- **Upstream Resolvers:** Cloudflare (1.1.1.1 / 1.0.0.1)
- **Performance:** Concurrent Forwarding enabled to minimize latency.

**Forwarders**

cloudflare-dns.com (1.1.1.1:853)  
cloudflare-dns.com (1.0.0.1:853)

**Quick Select**

Enter forwarder DNS Server IP addresses or URLs one below another in above text field or use the Quick Select list to select desired forwarder.

**Forwarder Protocol**

☐ DNS-over-UDP (default)  
☐ DNS-over-TCP  
☒ DNS-over-TLS  
☐ DNS-over-HTTPS  
☐ DNS-over-QUIC

Select a protocol that this DNS server must use to query the forwarders specified above.

**Concurrent Forwarding**

☒ Enable Concurrent Forwarding

Enable this option to allow querying two or more forwarders concurrently instead of sequentially querying them in their given order. The DNS server will automatically select forwarders (based on their average latency) to query and use the fastest response it receives from any of them. If none of the selected forwarders respond in time, the DNS server will similarly select forwarders from the remaining ones and queries them till all are tried before giving up.

**Forwarder Concurrency**

2 (valid range 1-10; default 2)

The number of concurrent requests that must be sent when Concurrent Forwarding is enabled for resolving a domain name.

Note! Forwarders are upstream DNS servers which this DNS Server must use to resolve domain names. If no forwarders are configured then the DNS server will use preconfigured ROOT HINTS to perform recursive resolution to resolve domain names.

Note! The `https` URL scheme supports only DNS-over-HTTPS/2 and DNS-over-HTTPS/1.1 protocols. For DNS-over-HTTPS/3, use `h3` URL scheme instead of `https` but note that there wont be any protocol fallback if the connection attempt fails.

Note! The DNS server uses Epsilon-Greedy machine learning algorithm and will automatically learn which of the forwarders are answering faster without errors and will use those forwarders most of the time.

Note! To customize the Quick Select drop down list, read the instructions given in the [www/json/readme.txt](#) file found in the installation folder.

[Help: Configuring DNS Server For Privacy & Security](#)

[Help: Configuring DNS-over-QUIC and HTTPS/3 For Technitium DNS Server](#)

**Forwarder Retries**

3 (valid range 1-10; default 3)

The total number of retries the forwarder or conditional forwarder resolver must do per upstream DNS server.

**Forwarder Timeout**

2000 milliseconds (valid range 1000-10000; default 2000)

The amount of time the forwarder or conditional forwarder resolver must wait between retries.

Save Settings

Flush Cache

Backup Settings

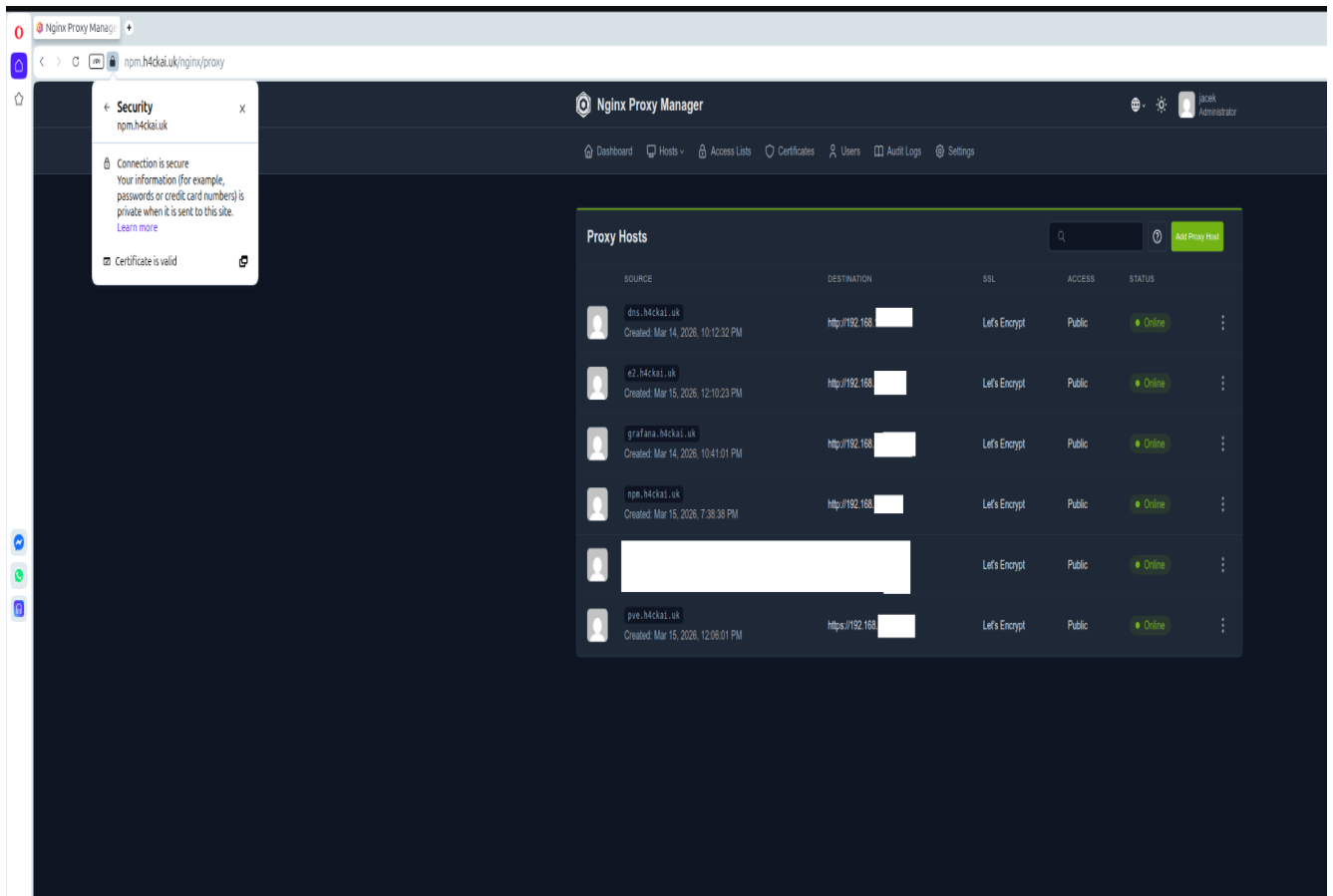
Restore Settings

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### 3.3 Reverse Proxy & SSL Termination

**Description:** Secure gateway for all internal web-based services.

- **Mechanism:** All subdomains (pve, dns, npm, wazuh) are protected by Let's Encrypt certificates.
- **Verification:** Browsers confirm "Connection is secure" with full certificate chain validation.



### 3.4 Traffic Validation & Security Audit

**Description:** Empirical verification of the implemented security measures.

- **Tool:** `tcpdump -n port 853`
- **Findings:** Live packet analysis confirms that all outbound DNS requests are encapsulated in TLS. No unencrypted traffic was detected on port 53.

```
root@technitiumdns:~# sudo tcpdump -n port 853
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
18:44:02.166342 IP 192.168.1.157.40092 > 1.1.1.1.853: Flags [S], seq 1852329004, win 64240, options [mss 1460,sackOK,TS val 863164204 ecr 0,nop,wscale 7], length 0
18:44:02.166373 IP 192.168.1.157.39984 > 1.0.0.1.853: Flags [S], seq 1597654610, win 64240, options [mss 1460,sackOK,TS val 2900950624 ecr 0,nop,wscale 7], length 0
18:44:02.177809 IP 1.1.1.1.853 > 192.168.1.157.40092: Flags [S.], seq 3893954618, ack 1852329005, win 65535, options [mss 1452,sackOK,TS val 1833422503 ecr 863164204,nop,wscale 13], length 0
18:44:02.177837 IP 192.168.1.157.40092 > 1.1.1.1.853: Flags [.] , ack 1, win 502, options [nop,nop,TS val 863164216 ecr 1833422503], length 0
18:44:02.178157 IP 192.168.1.157.40092 > 1.1.1.1.853: Flags [P.], seq 1:551, ack 1, win 502, options [nop,nop,TS val 863164216 ecr 1833422503], length 550
18:44:02.178626 IP 1.0.0.1.853 > 192.168.1.157.39984: Flags [S.], seq 61364713, ack 1597654611, win 65535, options [mss 1452,sackOK,TS val 3811048458 ecr 2900950624,nop,wscale 13], length 0
18:44:02.178638 IP 192.168.1.157.39984 > 1.0.0.1.853: Flags [.] , ack 1, win 502, options [nop,nop,TS val 2900950636 ecr 3811048458], length 0
18:44:02.178885 IP 192.168.1.157.39984 > 1.0.0.1.853: Flags [P.], seq 1:551, ack 1, win 502, options [nop,nop,TS val 2900950637 ecr 3811048458], length 550
18:44:02.190079 IP 1.1.1.1.853 > 192.168.1.157.40092: Flags [.] , ack 551, win 16, options [nop,nop,TS val 1833422515 ecr 863164216], length 0
18:44:02.190352 IP 1.1.1.1.853 > 192.168.1.157.40092: Flags [P.], seq 1:220, ack 551, win 16, options [nop,nop,TS val 1833422516 ecr 863164216], length 219
18:44:02.190355 IP 192.168.1.157.40092 > 1.1.1.1.853: Flags [.] , ack 220, win 501, options [nop,nop,TS val 863164228 ecr 1833422516], length 0
18:44:02.190595 IP 192.168.1.157.40092 > 1.1.1.1.853: Flags [P.], seq 551:631, ack 220, win 501, options [nop,nop,TS val 863164228 ecr 1833422516], length 80
18:44:02.193144 IP 1.0.0.1.853 > 192.168.1.157.39984: Flags [.] , ack 551, win 16, options [nop,nop,TS val 3811048473 ecr 2900950637], length 0
18:44:02.193334 IP 1.0.0.1.853 > 192.168.1.157.39984: Flags [P.], seq 1:220, ack 551, win 16, options [nop,nop,TS val 3811048473 ecr 2900950637], length 219
18:44:02.193338 IP 192.168.1.157.39984 > 1.0.0.1.853: Flags [.] , ack 220, win 501, options [nop,nop,TS val 2900950651 ecr 3811048473], length 0
18:44:02.193451 IP 192.168.1.157.39984 > 1.0.0.1.853: Flags [P.], seq 551:631, ack 220, win 501, options [nop,nop,TS val 2900950651 ecr 3811048473], length 80
18:44:02.220385 IP 192.168.1.157.39984 > 1.0.0.1.853: Flags [P.], seq 631:713, ack 220, win 501, options [nop,nop,TS val 2900950678 ecr 3811048473], length 82
18:44:02.221858 IP 192.168.1.157.40092 > 1.1.1.1.853: Flags [P.], seq 631:713, ack 220, win 501, options [nop,nop,TS val 863164260 ecr 1833422516], length 82
18:44:02.232202 IP 1.1.1.1.853 > 192.168.1.157.40092: Flags [.] , ack 713, win 16, options [nop,nop,TS val 1833422557 ecr 863164228], length 0
18:44:02.232697 IP 1.0.0.1.853 > 192.168.1.157.39984: Flags [.] , ack 713, win 16, options [nop,nop,TS val 3811048512 ecr 2900950651], length 0
18:44:02.234203 IP 1.0.0.1.853 > 192.168.1.157.39984: Flags [P.], seq 220:1162, ack 713, win 16, options [nop,nop,TS val 3811048514 ecr 2900950651], length 942
18:44:02.235966 IP 1.1.1.1.853 > 192.168.1.157.40092: Flags [P.], seq 220:1162, ack 713, win 16, options [nop,nop,TS val 1833422561 ecr 863164228], length 942
18:44:02.274910 IP 192.168.1.157.39984 > 1.0.0.1.853: Flags [.] , ack 1162, win 494, options [nop,nop,TS val 2900950733 ecr 3811048514], length 0
18:44:02.276906 IP 192.168.1.157.40092 > 1.1.1.1.853: Flags [.] , ack 1162, win 494, options [nop,nop,TS val 863164315 ecr 1833422561], length 0
18:44:02.731892 IP 192.168.1.157.40092 > 1.1.1.1.853: Flags [P.], seq 713:781, ack 1162, win 494, options [nop,nop,TS val 863164770 ecr 1833422561], length 68
18:44:02.731922 IP 192.168.1.157.39984 > 1.0.0.1.853: Flags [P.], seq 713:781, ack 1162, win 494, options [nop,nop,TS val 2900951190 ecr 3811048514], length 68
18:44:02.732055 IP 192.168.1.157.39984 > 1.0.0.1.853: Flags [P.], seq 781:849, ack 1162, win 494, options [nop,nop,TS val 2900951190 ecr 3811048514], length 68
18:44:02.732055 IP 192.168.1.157.40092 > 1.1.1.1.853: Flags [P.], seq 781:849, ack 1162, win 494, options [nop,nop,TS val 863164770 ecr 1833422561], length 68
18:44:02.743512 IP 1.1.1.1.853 > 192.168.1.157.40092: Flags [P.], seq 1162:1654, ack 781, win 16, options [nop,nop,TS val 1833423069 ecr 863164770], length 492
18:44:02.743524 IP 192.168.1.157.40092 > 1.1.1.1.853: Flags [.] , ack 1654, win 491, options [nop,nop,TS val 863164781 ecr 1833423069], length 0
18:44:02.744546 IP 1.1.1.1.853 > 192.168.1.157.40092: Flags [P.], seq 1654:2146, ack 849, win 16, options [nop,nop,TS val 1833423070 ecr 863164770], length 492
18:44:02.744563 IP 192.168.1.157.40092 > 1.1.1.1.853: Flags [.] , ack 2146, win 488, options [nop,nop,TS val 863164782 ecr 1833423070], length 0
18:44:02.745599 IP 1.0.0.1.853 > 192.168.1.157.39984: Flags [.] , ack 849, win 16, options [nop,nop,TS val 3811049026 ecr 2900951190], length 0
18:44:02.746803 IP 1.0.0.1.853 > 192.168.1.157.39984: Flags [P.], seq 1162:2124, ack 849, win 16, options [nop,nop,TS val 3811049027 ecr 2900951190], length 962
18:44:02.746831 IP 192.168.1.157.39984 > 1.0.0.1.853: Flags [.] , ack 2124, win 487, options [nop,nop,TS val 2900951205 ecr 3811049027], length 0
18:44:02.852423 IP 192.168.1.157.40092 > 1.1.1.1.853: Flags [P.], seq 849:917, ack 2146, win 488, options [nop,nop,TS val 863164890 ecr 1833423070], length 68
```

## 4. Conclusion

The infrastructure successfully mitigates common network-level threats, including DNS Spoofing and Man-in-the-Middle attacks. The current setup provides a robust foundation for further security implementations, such as SIEM (Wazuh) integration.

### End of Report